

```
/* C Program to implement Linear Search Algorithm recursively */
```

```
#include <stdio.h>
```

```
int RecursiveLS(int arr[], int value, int index, int n)
```

```
{
```

```
    int pos = 0;
```

```
    if(index >= n)
```

```
    {
```

```
        return 0;
```

```
    }
```

```
    else if (arr[index] == value)
```

```
    {
```

```
        pos = index + 1;
```

```
        return pos;
```

```
    }
```

```
    else
```

```
    {
```

```
        return RecursiveLS(arr, value, index+1, n);
```

```
    }
```

```
    return pos;
```

```
}
```

```
int main()

{

    int n, value, pos, m = 0, arr[100];

    printf("Enter the total elements in the array: ");

    scanf("%d", &n);


    printf("Enter the array elements:\n");

    for (int i = 0; i < n; i++)

    {

        scanf("%d", &arr[i]);

    }


    printf("Enter the element to search: ");

    scanf("%d", &value);


    pos = RecursiveLS(arr, value, 0, n);

    if (pos != 0)

    {

        printf("Element found at pos %d ", pos);

    }

    else

    {

        printf("Element not found");

    }

    return 0;
```

}